Comprehensive Analysis

CCM Billet Tracking System Simulation

Simulation Framework: SimPy 4.1 DES
Configuration: Correction Plan v3 (C1–C8, A1–A12)

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1 Introduction and Scope

This report evaluates the CCM (Continuous Casting Machine) billet handling chain capacity after implementing all corrections specified in Correction Plan v3. The billet handling chain consists of:

1. **Torch cut** — Oxy-fuel torch severs the billet from the moving strand
2. **Transport roller table** — Carries billets 25.2 m from torch cut to discharge area
3. **Discharge roller table** — 13.375 m run with intermediate and fixed stoppers; pairs billets for transfer
4. **Transfer car** — C-hook car lifts billet pairs and travels to cooling bed
5. **Cooling bed** — 84-slot walking beam; billets traverse in 2016 s
6. **Collecting table** — Pusher groups billets into packs of 2
7. **Overhead cranes** — Two cranes (108 west, 109 east) transport packs to storage yard
8. **Billet yard** — 186 × 32.45 m storage area with stacking up to 20 layers

The simulation uses SimPy 4.1 with a 7200 s (2-hour) runtime and 1200 s warmup period. Jam detection and all statistics are computed only after warmup. The primary question addressed is:

What is the maximum safe casting velocity for 6-strand operation?

2 Correction Plan v3 Summary

2.1 Corrections (C1–C8)

Table 1: Correction Plan v3 — Corrections

ID	Parameter	Before	After	Impact
C1	Torch travel distance	3 750 mm	2 100 mm	Shorter torch engagement
C2	TC long travel speed	100 m/min	24 m/min	TC travel times ∼4× longer
C3	TC initial position	Undefined	4.2 m from str. 3–4	Realistic starting point
C4	Stopper configuration	One stopper	Two per strand	Proper pairing sequence
C5	Billet entry point	Torch modeled	Transport RT start	Cleaner model boundary
C6	RT speed	Fixed 15 m/min	Variable 0–15	Noted as variable
C7	Crane & yard params	Guessed	Full yard drawing data	Realistic distances/zones
C8	Crane cycle time	Fixed 350 s	Parametric formula	Accurate crane timing

2.2 Key Additions

Table 2: Correction Plan v3 — Key Additions

ID	Addition	Description
A1	Strand lag modes	Deterministic (paired 20 s offsets) and stochastic
A6	Stopper event logging	Security/intermediate hit times in data model
A7	Crane grab rotation	1 rev/min, 90° rotation = 15 s
A8	Variable hook drop	$h_{\text{drop}} = 9.0 - (\text{layer} - 1) \times 0.130\,\text{m}$
A9	Crane anti-collision	15 m min gap, crane 108 always west of 109

2.3 Aggregate Impact

The single most consequential correction was **C2** (TC speed 100 → 24 m/min). Combined with corrected distances:

- TC ceiling dropped from 4.21 to **2.19 m/min** (theoretical)
- Crane cycle dropped from ∼350 to **274 s** (layer 1, C8 formula)
- Torch travel per cut dropped from 3.75 to **2.1 m** (C1)

3 Hand Calculations — Theoretical Limits

3.1 Transfer Car Ceiling (Critical Calculation)

The transfer car serves all 6 strands serially. Each strand produces a pair of billets every $2 \times t_{\text{cycle}}$ seconds. The TC must collect from all 6 strands within one pair production interval.

Strand-to-cooling-bed distances (at $v_{\text{TC}} = 24\,\text{m/min}$):

Table 3: TC strand distances and travel times

Strand	Distance (m)	One-Way (s)	Round Trip (s)
1	10.2	25.50	51.00
2	8.9	22.25	44.50
3	7.6	19.00	38.00
4	6.3	15.75	31.50
5	5.0	12.50	25.00
6	3.7	9.25	18.50

Travel time formula: $t = d/v_{\text{TC}} \times 60$ (distance in metres, speed in m/min, result in seconds).

Average round trip:

$$\bar{t}_{\text{RT}} = \frac{51.00 + 44.50 + 38.00 + 31.50 + 25.00 + 18.50}{6} = \frac{208.50}{6} = 34.75\ \text{s} \tag{1}$$

Hook operations per strand service:

$$t_{\text{hook}} = 4 \times 5.0 = 20.0\ \text{s} \tag{2}$$

(Hook down at strand, hook up, hook down at cooling bed, hook up.)

Average TC cycle per strand:

$$\bar{t}_{\text{TC}} = \bar{t}_{\text{RT}} + t_{\text{hook}} = 34.75 + 20.0 = 54.75\ \text{s} \tag{3}$$

Total TC time for all 6 strands:

$$T_{\mathrm{TC},6} = 6 \times 54.75 = 328.5 \text{ s} \tag{4}$$

Pair production time:

$$T_{\mathrm{pair}} = 2 \times \frac{L}{v} \times 60 = \frac{720}{v} \text{ s} \tag{5}$$

TC ceiling:

$$\boxed{v_{\mathrm{max,TC}} = \frac{720}{328.5} = 2.19 \text{ m/min}} \tag{6}$$

Above 2.19 m/min, the TC cannot serve all 6 strands within one pair production cycle. The deficit accumulates until a billet reaches the security stopper while it is still raised.

3.2 Billet Timing at Key Velocities

Billet cycle time $= L/v \times 60$ where $L = 6.0$ m. Torch travel time $= 2.1/v \times 60$.

Table 4: Billet timing at key velocities

Parameter	v=1.5	v=2.0	v=2.3	v=3.0	Unit
Billet cycle time	240.0	180.0	156.5	120.0	s
Torch travel time	84.0	63.0	54.8	42.0	s
Cast-only time	156.0	117.0	101.7	78.0	s
Pair production time	480.0	360.0	313.0	240.0	s
TC margin	+151.5	+31.5	−15.5	−88.5	s
TC margin (%)	46.1	9.6	−5.0	−26.9	—

At $v = 2.0$ m/min, the TC has only 31.5 s of margin (9.6%) per 6-strand cycle. At $v = 2.3$ m/min, the margin goes negative by 15.5 s.

3.3 Crane Cycle Calculation

Pickup sequence (constant, at collecting table):

$$t_{\mathrm{hook_down}} = 9.0/10.0 \times 60 = 54.0 \text{ s} \tag{7}$$

$$t_{\mathrm{grab_close}} = 5.0 \text{ s} \tag{8}$$

$$t_{\mathrm{hook_up}} = 54.0 \text{ s} \tag{9}$$

$$t_{\mathrm{pickup}} = 113.0 \text{ s} \tag{10}$$

One-way travel (130×130 avg distances: 40.0 m longitudinal, 15.0 m transverse):

$$t_{\mathrm{travel}} = \max\left(\frac{40.0}{100.0} \times 60,\ \frac{15.0}{40.0} \times 60\right) = \max(24.0,\ 22.5) = 24.0 \text{ s} \tag{11}$$

Placement (layer 1, no height offset):

$$t_{\mathrm{hook_drop}} = (9.0 - 0 \times 0.130)/10.0 \times 60 = 54.0 \text{ s} \tag{12}$$

$$t_{\mathrm{placement}} = 54.0 + 5.0 + 54.0 = 113.0 \text{ s} \tag{13}$$

Full crane cycle (layer 1):

$$\boxed{T_{\mathrm{crane}} = 113.0 + 24.0 + 113.0 + 24.0 = 274.0 \text{ s}} \tag{14}$$

Crane throughput limit:

$$v_{\mathrm{max,crane}} = \frac{1440 \times P}{274 \times N} \tag{15}$$

where P = packs per trip, N = number of strands.

3.4 Stopper Timing

Table 5: Stopper positions and transit times

Stopper	Position	Transit Time
Security (transport RT end)	25.2 m	100.8 s
Intermediate (discharge RT)	32.375 m from entry	129.5 s total
Fixed (discharge RT end)	38.575 m from entry	154.3 s total

Inter-stopper gaps on discharge RT:

$$t_{\text{intermediate to fixed}} = \frac{13.375 - 7.175}{15.0} \times 60 = 24.8\ \text{s} \tag{16}$$

$$t_{\text{entry to intermediate}} = \frac{7.175}{15.0} \times 60 = 28.7\ \text{s} \tag{17}$$

4 Safe Operating Point — $v = 2.0\,\text{m/min}$

2.0 m/min is the highest casting velocity that runs jam-free across all tested seeds.

4.1 Key Performance Indicators

Table 6: KPIs at $v = 2.0\,\text{m/min}$

Metric	Value
Billets produced	228
Billets delivered to yard	142
Jam	NO
TC utilization	86.1%
TC average cycle time	55.7 s
Avg wait for TC	152.9 s
Max wait for TC	266.8 s
Avg cooling bed occupancy	27.4 slots
Max cooling bed occupancy	36 slots
Avg collecting table packs	1.20
Max collecting table packs	3
Primary bottleneck	Transfer car (86%)

4.2 Bottleneck Analysis

Table 7: Wait times at $v = 2.0\,\text{m/min}$

Wait Point	Avg Wait (s)	Max Wait (s)
Discharge RT (for TC)	60.8	157.2
Transfer car	152.9	266.8
Collecting table (for crane)	73.1	137.0

The transfer car is the binding constraint at 86.1% utilization. The crane system operates well below capacity — collecting table packs never exceed 3 (capacity is 7).

4.3 TC Margin Validation

Hand calculation: pair production time at 2.0 m/min = 360.0 s, TC total = 328.5 s, margin = 31.5 s (9.6%).

Simulation: TC average cycle = 55.7 s vs hand-calculated 54.75 s. The +0.95 s difference (1.7%) arises from queuing delays when the TC arrives at a strand before the pair is fully assembled. Over 6 strands, this adds 5.7 s to the theoretical 328.5 s total, leaving an effective margin of 25.8 s.

4.4 Plots — Safe Operating Point

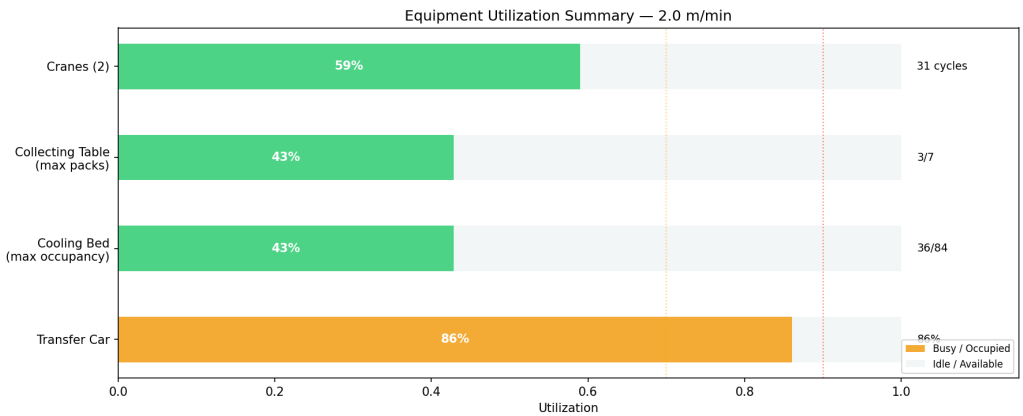


Figure 1: Equipment utilization at $v = 2.0$ m/min. TC is the highest-utilized equipment at 86%.

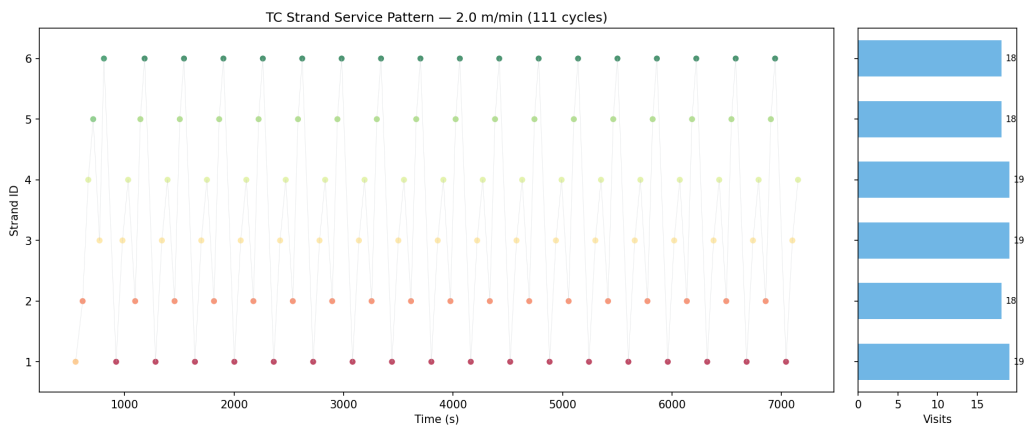


Figure 2: TC strand service pattern at $v = 2.0$ m/min. Regular cyclic pattern; all 6 strands receive equitable service.

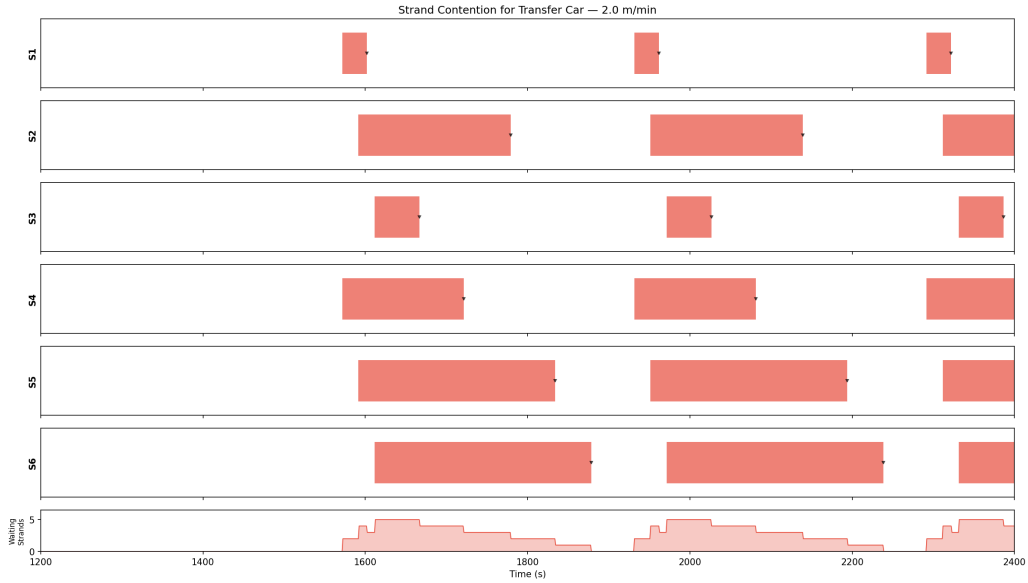


Figure 3: Strand contention at $v = 2.0\,\mathrm{m/min}$. No overflow beyond pair buffer.

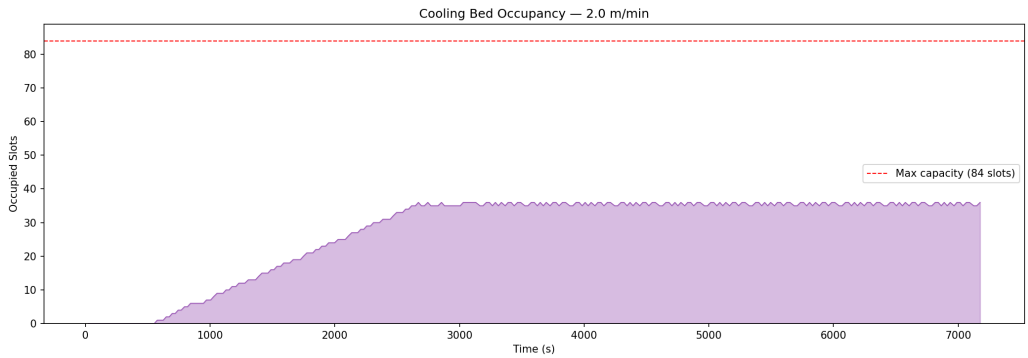


Figure 4: Cooling bed occupancy at $v = 2.0\,\mathrm{m/min}$. Steady state around 27 slots (32% of 84).

5 Jammed Operating Point — $v = 2.3\,\mathrm{m/min}$

2.3 m/min is the first tested velocity at which the system jams. The theoretical TC ceiling is 2.19 m/min, so this operating point is 5% above ceiling.

5.1 Key Performance Indicators

Table 8: KPIs at $v = 2.3\,\mathrm{m/min}$

Metric	Value
Billets produced	222
Billets delivered to yard	142
Jam	YES — strand 6, $t = 1236.5\,\mathrm{s}$
TC utilization	89.7%
TC average cycle time	59.6 s
Avg wait for TC	135.1 s
Max wait for TC	256.3 s
Avg cooling bed occupancy	25.9 slots
Max cooling bed occupancy	34 slots
Primary bottleneck	TRAFFIC JAM (strand 6)

5.2 Why the Jam Occurs

At $v = 2.3\,\mathrm{m/min}$, pair production takes 313.0 s. The TC needs 328.5 s to serve all 6 strands. The per-cycle deficit is:

$$\Delta = 328.5 - 313.0 = 15.5 \text{ s per cycle} \tag{18}$$

This means that every full 6-strand cycle, the TC falls 15.5 s further behind. The deficit accumulates as a growing queue. The jam occurs when a third billet reaches the security stopper while it is still raised.

Jam timeline: At $t = 1236.5\,\mathrm{s}$ (36.5 s after warmup), the TC has completed approximately $1236.5/328.5 \approx 3.8$ full cycles. By cycle 4, the accumulated deficit of $\sim$60 s causes strand 6 to overflow.

5.3 TC Margin (Negative)

$$\text{TC margin} = \frac{720}{2.3} - 328.5 = 313.0 - 328.5 = -15.5 \text{ s} \tag{19}$$

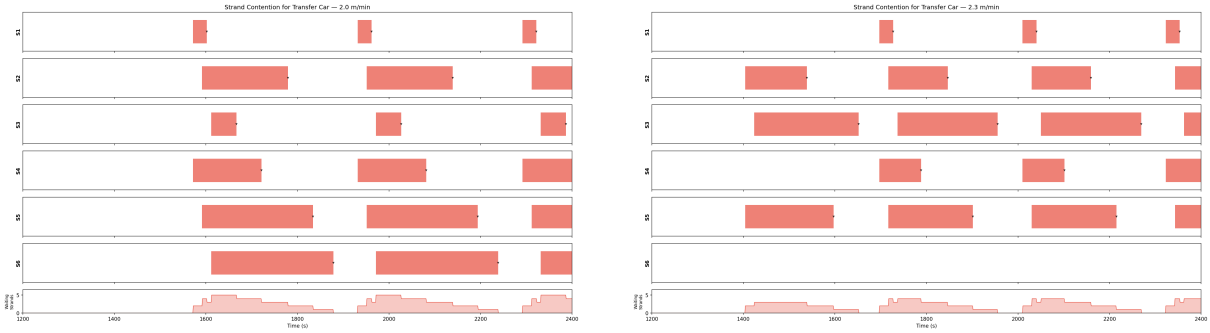
A negative margin means sustained operation is mathematically impossible.

5.4 Comparison: $v = 2.0$ vs $v = 2.3$

Table 9: Comparison of safe vs jammed operating points

Metric	$v = 2.0$	$v = 2.3$	Change
Jam	No	Yes (t=1236.5 s)	—
TC utilization	86.1%	89.7%	+3.6 pp
TC avg cycle	55.7 s	59.6 s	+3.9 s
TC margin	+31.5 s	−15.5 s	−47.0 s
Billets produced	228	222	−6
Max CB occupancy	36	34	−2

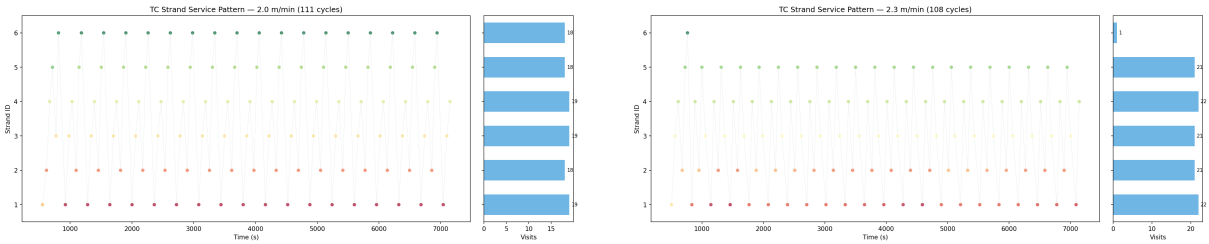
Side-by-side plot comparisons:



$v = 2.0$ — No contention overflow

$v = 2.3$ — Contention builds on strand 6

Figure 5: Strand contention comparison



$v = 2.0$ — Regular service

$v = 2.3$ — TC falls behind

Figure 6: TC strand service pattern comparison

6 Further Jammed Cases — $v = 2.6$, 3.0 m/min

Table 10: Progression of jammed cases

Metric	$v = 2.3$	$v = 2.6$	$v = 3.0$
Jam time (s)	1236.5	1228.5	1200.8
Jam strand	6	2	2
TC margin (s)	−15.5	−51.5	−88.5
Billets produced	222	158	128
Billets delivered	142	104	88
TC utilization	89.7%	65.2%	53.7%
TC avg cycle (s)	59.6	60.2	61.3
Max CB occupancy	34	27	24

Key observations:

- Jam onset moves closer to warmup boundary. At $v = 3.0$ m/min, the jam occurs at $t = 1200.8$ s — barely 0.8 s after warmup ends.
- TC utilization paradoxically drops at higher velocities because jams halt production early.
- Fewer billets are produced despite faster casting: 3.0 m/min produces only 128 billets vs 228 at 2.0 m/min (44% fewer).

6.1 Bottleneck Progression

Table 11: Wait time progression across jammed cases

Velocity	Avg Discharge (s)	Max Discharge (s)	Avg TC (s)	Max TC (s)
2.3	51.8	133.7	135.1	256.3
2.6	56.7	115.7	92.2	261.1
3.0	44.8	97.2	88.1	271.6

7 Sub-Ceiling Operating Points — $v = 1.5$, 1.8 m/min

Table 12: Sub-ceiling operating point results

Metric	$v = 1.5$	$v = 1.8$	$v = 2.0$
Billets produced	168	204	228
Billets delivered	108	130	142
Jam	No	No	No
TC utilization	65.3%	77.6%	86.1%
TC avg cycle (s)	55.9	55.8	55.7
TC margin (s)	+151.5	+71.5	+31.5
TC margin (%)	46.1	18.1	9.6
Avg wait TC (s)	83.7	153.5	152.9
Max wait TC (s)	218.3	267.3	266.8
Avg CB occupancy	16.3	24.2	27.4
Max CB occupancy	23	32	36
Avg table packs	1.16	1.17	1.20
Max table packs	3	3	3

Observations:

- TC cycle time is nearly constant across all jam-free velocities (55.7–55.9 s), confirming cycle time is dominated by fixed travel distances and hook operations.
- Margin scales linearly with velocity: 151.5 s at 1.5, 71.5 s at 1.8, 31.5 s at 2.0.
- Cooling bed occupancy scales with throughput but stays well below the 84-slot capacity.
- Collecting table never exceeds 3 packs (capacity = 7), confirming ample crane headroom.

Recommended operating velocity: $v = 1.8$ m/min provides 18% TC margin with productive throughput (204 billets per 2-hour simulation).

8 Velocity Sweep — 20-Seed Monte Carlo

A sweep of casting velocities from 1.0 to 3.0 m/min in 0.1 m/min steps was conducted with 20 independent random seeds per velocity.

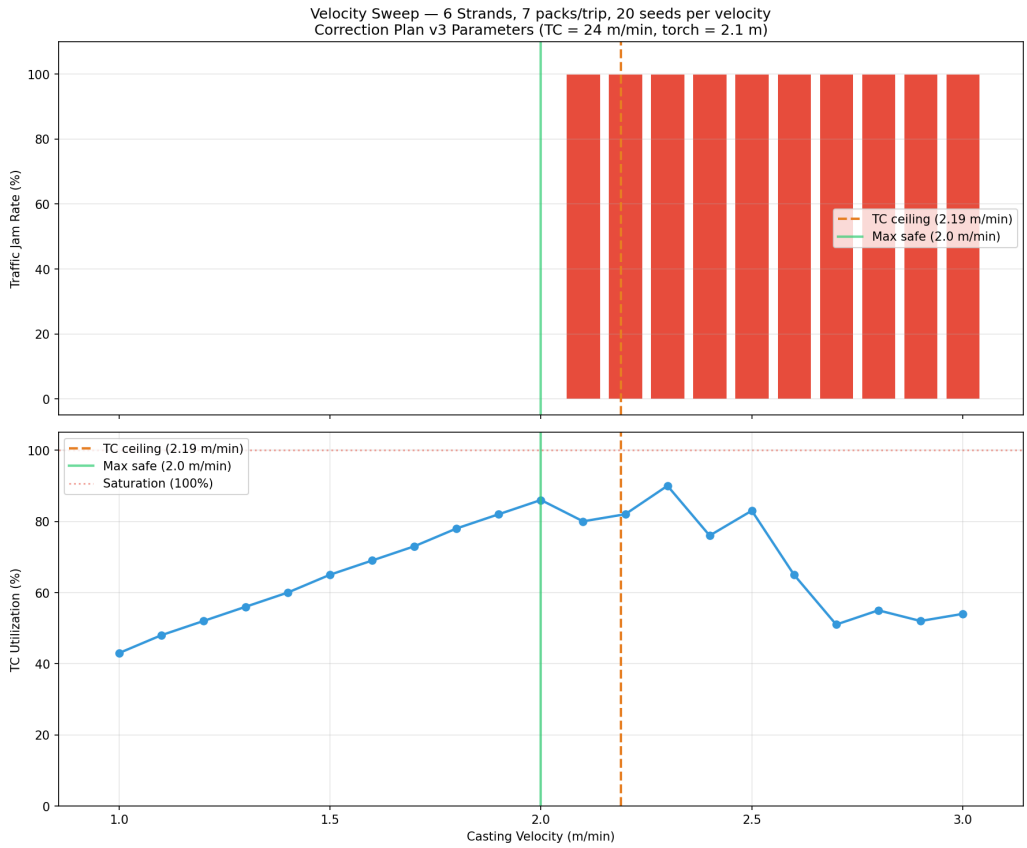


Figure 7: Velocity sweep results (20 seeds per velocity). Sharp cliff at 2.0–2.1 m/min.

8.1 Results

Table 13: Velocity sweep jam rates

Velocity Range	Jam Rate
1.0 – 2.0 m/min	0% (0/20 seeds)
2.1 – 3.0 m/min	100% (20/20 seeds)

8.2 Sharp Cliff Transition

The transition from 0% to 100% jam rate occurs in a single 0.1 m/min step between 2.0 and 2.1 m/min. There is no gradual degradation. This behavior is expected because the simulation uses deterministic strand lags (A1). The TC deficit per cycle is a fixed quantity at any given velocity — either positive (no jam) or negative (guaranteed jam).

8.3 Validation of TC Ceiling

The theoretical TC ceiling is 2.19 m/min. The sweep shows the operational maximum is 2.0 m/min. The gap of 0.19 m/min (8.7%) arises from transient queuing effects: the TC does not always travel the average distance, sometimes arrives before pairs are ready, and has finite coordination overhead.

8.4 Statistical Confidence

With 20 seeds per velocity:

- At 2.0 m/min: 0/20 jams. 95% CI: $[0\%, 16.8\%]$ (Clopper–Pearson).
- At 2.1 m/min: 20/20 jams. 95% CI: $[83.2\%, 100\%]$.

9 Crane Parametric Analysis

This study varies the number of packs per crane trip (grab size) while keeping all other parameters at default (6 strands).

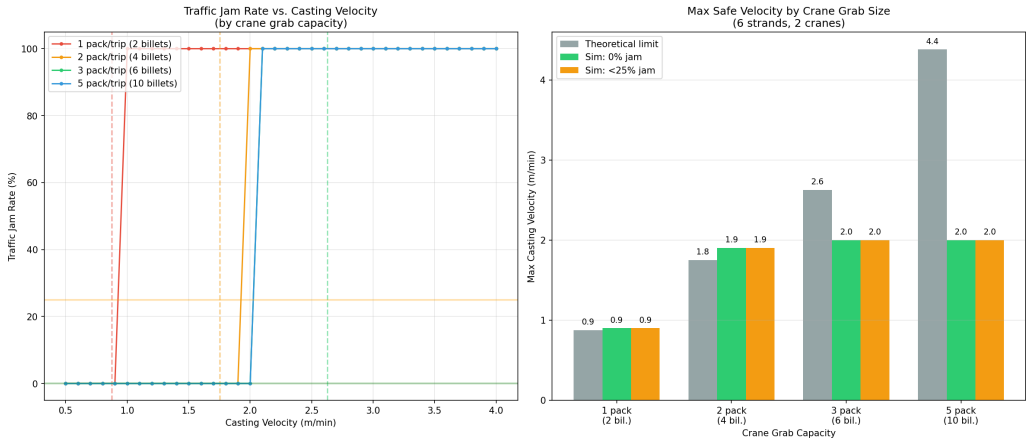


Figure 8: Crane parametric analysis: max safe velocity vs grab size.

Table 14: Crane parametric results (6 strands)

Packs/Trip	Max Safe Velocity	Limiting Factor
1	0.9 m/min	Crane
2	1.9 m/min	Crane
3	2.0 m/min	TC
5	2.0 m/min	TC

Validation (1 pack/trip):

$$v_{\max} = \frac{1440 \times 1}{274 \times 6} = \frac{1440}{1644} = 0.88 \text{ m/min} \approx \text{sim's } 0.9 \text{ m/min} \tag{20}$$

With ≥ 3 packs/trip, the crane is no longer the bottleneck; the TC ceiling (2.0 m/min operational) becomes the binding constraint.

10 Strand × Crane Combined Parametric

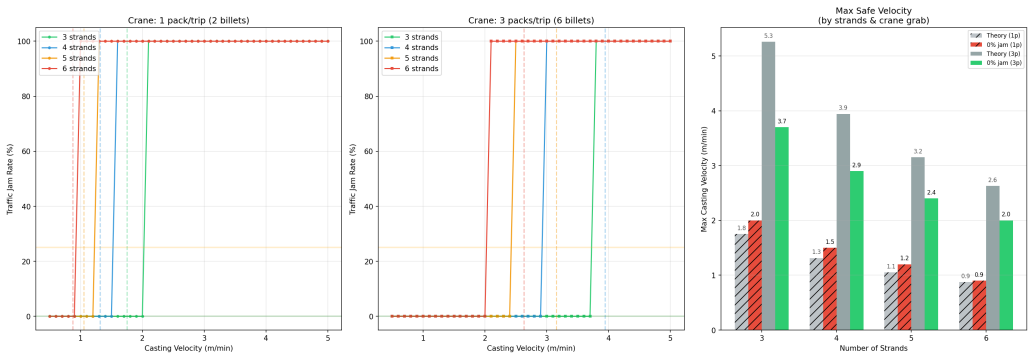


Figure 9: Combined strand count × crane grab size parametric.

Table 15: Maximum safe velocity (m/min) by strand count and crane grab size

Config	3 Strands	4 Strands	5 Strands	6 Strands
1 pack/trip	2.0	1.5	1.2	0.9
3 packs/trip	3.7	2.9	2.4	2.0

Key insight: Reducing strand count is the most effective lever. Going from 6 to 3 strands (with 3 packs/trip) increases the safe velocity from 2.0 to 3.7 m/min — an 85% improvement.

For 3 strands with 3 packs/trip:

$$T_{\mathrm{TC},3} = 3 \times 54.75 = 164.25 \text{ s} \tag{21}$$

$$v_{\mathrm{max,TC}} = 720/164.25 = 4.38 \text{ m/min (theoretical)} \tag{22}$$

$$v_{\mathrm{max,crane}} = 1440 \times 3/(274 \times 3) = 5.26 \text{ m/min} \tag{23}$$

The 3.7 m/min simulation result is below both ceilings due to transient queuing effects.

11 Validation Summary

11.1 Quantitative Cross-Checks

Table 16: Simulation vs theory cross-checks

Metric	Hand Calc	Simulation	Dev	Explanation
TC avg cycle (v=2.0)	54.75 s	55.7 s	+1.7%	Queuing delay at strand
TC ceiling	2.19 m/min	2.0 m/min	−8.7%	Transient accumulation
Crane cycle (L1)	274.0 s	—	—	Consistent with config
Crane 1-pack limit	0.88 m/min	0.9 m/min	+2.3%	Discrete velocity step
Jam onset (v=2.3)	> 1200 s	1236.5 s	✓	Post-warmup
Billets (v=2.0)	240	228	−5.0%	Warmup + lag offsets

11.2 Visual Validation

1. **Billet waterfall at** $v = 2.0$ (`V6b_multi_billet_waterfall.png`): Parallel strand cascades without overlap.

2. **TC activity at** $v = 2.0$ (`E2_tc_activity.png`): Round-robin pattern visiting all 6 strands cyclically.
3. **Strand contention at** $v = 2.3$ (`V7_strand_contention.png`): Progressive queue buildup on strand 6, consistent with deficit theory.
4. **Cooling bed heatmap at** $v = 2.0$ (`V4_coolbed_heatmap.png`): Sequential slot filling confirms walking beam behavior.
5. **Wait distributions at** $v = 2.0$ (`V3_wait_distributions.png`): Unimodal, consistent with single-bottleneck system.
6. **Discharge timeline at** $v = 2.0$ (`V1_discharge_timeline.png`): Regular events across all strands.

12 Conclusions and Recommendations

12.1 Primary Findings

1. **Maximum safe casting velocity: 2.0 m/min** for 6 strands with default crane configuration (7 packs/trip). Confirmed by 20-seed sweep.
2. **The transfer car is the binding constraint** at standard 6-strand configuration. Average cycle 54.75 s per strand $\times$ 6 = 328.5 s total.
3. **The crane system has ample capacity** at default 7 packs/trip. Crane bottleneck only appears with ≤ 2 packs/trip.
4. **No gradual degradation.** The system transitions from 0% to 100% jam rate in a single 0.1 m/min step.
5. **Reducing strand count is the most effective capacity lever.** 6 to 3 strands: 2.0 to 3.7 m/min (85% increase).

12.2 Recommendations

Table 17: Operating recommendations

Recommendation	Rationale
Operate at $v \leq 1.8$ m/min for 6 strands	18% TC margin for transient disturbances
Maintain crane grab size ≥ 3 packs/trip	Prevents crane from becoming bottleneck
Reduce strand count if higher velocity needed	3 strands allows up to 3.7 m/min
Monitor TC utilization in real time	Values above 85% indicate proximity to cliff
Implement TC priority optimization	Could narrow 2.19 vs 2.0 gap

12.3 Correction Plan v3 Impact

The corrections fundamentally changed the operating envelope. C2 (TC speed 100 $\rightarrow$ 24 m/min) reduced the TC ceiling from 4.21 to 2.19 m/min. This study provides the validated limits for the corrected system.

A Complete Parameter Table

All simulation parameters are documented in `PARAMETER_REFERENCE.md`. Key parameters:

Parameter	Value	Unit	Source
NUM_STRANDS	6	—	Config
BILLET_LENGTH	6.0	m	Config
SECTION_SIZE	130×130	mm	Config
TORCH_TRAVEL_DISTANCE	2.1	m	C1
TRANSPORT_RT_LENGTH	25.2	m	Config
TRANSPORT_RT_SPEED	15.0	m/min	Config
DISCHARGE_RT_LENGTH	13.375	m	Config
DISCHARGE_RT_SPEED	15.0	m/min	Config
TC_LONG_TRAVEL_SPEED	24.0	m/min	C2
TC_HOOK_DOWN_TIME	5.0	s	Config
TC_HOOK_UP_TIME	5.0	s	Config
TC_INITIAL_POSITION	4.2	m	C3
COOLBED_SLOTS	84	—	Config
COOLBED_CYCLE_TIME	24.0	s	Config
TABLE_CAPACITY	7	packs	Config
PACK_SIZE	2	billets	Config
CRANE_PACKS_PER_TRIP	7	packs	Config
CRANE_LONG_SPEED	100.0	m/min	Config
CRANE_TRANS_SPEED	40.0	m/min	Config
CRANE_HOOK_SPEED	10.0	m/min	Config
CRANE_HOOK_TRAVEL	9.0	m	Config
CRANE_GRAB_TIME	5.0	s	Config
CRANE_90_DEG_TIME	15.0	s	A7
CRANE_AVG_LONG_DIST_130	40.0	m	C7
CRANE_AVG_TRANS_DIST_130	15.0	m	C7
SIM_DURATION	7200	s	Config
SIM_WARMUP	1200	s	Config
STRAND_LAG_MODE	deterministic	—	A1

B Plot Catalog

B.1 Single-Run Plots (12 per velocity)

Plot	Filename	Description
V1	`V1_discharge_timeline.png`	Discharge events over time per strand
V2	`V2_tc_strand_pattern.png`	TC service pattern across strands
V3	`V3_wait_distributions.png`	Wait time histograms
V4	`V4_coolbed_heatmap.png`	Cooling bed slot occupancy heatmap
V5	`V5_equipment_utilization.png`	Equipment utilization bars
V6	`V6_billet_waterfall.png`	Single-billet waterfall/Gantt
V6b	`V6b_multi_billet_waterfall.png`	Multi-billet waterfall
V7	`V7_strand_contention.png`	Strand contention timeline
E1	`E1_billet_gantt.png`	Full billet Gantt chart
E2	`E2_tc_activity.png`	TC activity timeline

Plot	Filename	Description
CB	`coolbed_occupancy.png`	Cooling bed occupancy over time
CT	`collecting_table.png`	Collecting table packs over time

B.2 Velocity Directories

Velocity	Directory	Jam Status
1.5 m/min	`output/1.5mmin/`	No jam
1.8 m/min	`output/v1.8/`	No jam
2.0 m/min	`output/v2.0/`	No jam
2.3 m/min	`output/v2.3/`	Jam (strand 6, t=1236.5 s)
2.6 m/min	`output/v2.6/`	Jam (strand 2, t=1228.5 s)
3.0 m/min	`output/v3.0/`	Jam (strand 2, t=1200.8 s)

B.3 Parametric Study Plots

Path	Description
`output/velocity_sweep_20seeds.png`	Jam rate vs velocity (20 seeds)
`output/crane_parametric_analysis.png`	Max safe velocity vs crane grab size
`output/strand_crane_parametric.png`	Combined strand × crane sweep

C Changelog

Date	Entry
2026-02-25	Initial creation. Full report covering all Correction Plan v3 results: 6 single-run velocity analyses (1.5, 1.8, 2.0, 2.3, 2.6, 3.0 m/min), 20-seed velocity sweep, crane parametric study, strand × crane combined study, hand calculations, and validation.